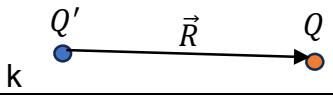
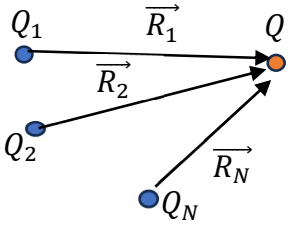
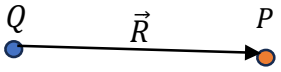
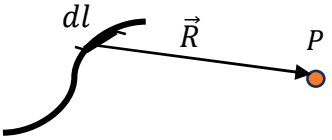
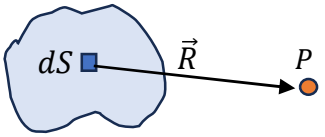
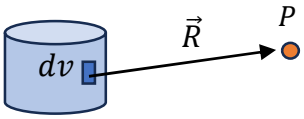
6. ELECTROSTATIC

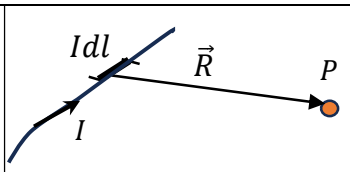
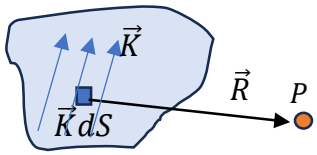
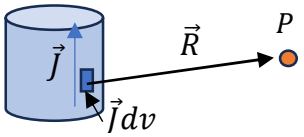
<table>
<tr><td rowspan="2">Coulomb's law
Electric force

$\vec{F}\ [N]$</td><td>$\vec{F} = \frac{QQ'}{4\pi\varepsilon}\frac{\vec{R}}{R^3} = \frac{QQ'}{4\pi\varepsilon}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td> 
</td></tr>
<tr><td>$\vec{F} = \frac{Q}{4\pi\varepsilon}\sum_{k=1}^{N}\frac{Q_k\overrightarrow{R_k}}{{R_k}^3} = \frac{Q}{4\pi\varepsilon}\sum_{k=1}^{N}\frac{Q_k(\vec{r}-\overrightarrow{r_k})}{\left|\vec{r}-\overrightarrow{r_k}\right|^3}$</td><td> 
</td></tr>
<tr><td rowspan="4">Electric field intensity

$\vec{E}$</td><td>Electric field due to point charge
$\vec{E} = \frac{\vec{F}}{Q} = \frac{Q'}{4\pi\varepsilon}\frac{\vec{R}}{R^3} = \frac{Q'}{4\pi\varepsilon}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td> 
</td></tr>
<tr><td>Electric field due to line charge
$\vec{E} = \int_L \frac{\rho_L dl}{4\pi\varepsilon}\frac{\vec{R}}{R^3} = \int_L \frac{\rho_L dl}{4\pi\varepsilon}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td> 
</td></tr>
<tr><td>Electric field due to surface charge
$\vec{E} = \int_S \frac{\rho_s ds}{4\pi\varepsilon}\frac{\vec{R}}{R^3} = \int_S \frac{\rho_s ds}{4\pi\varepsilon}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td> 
</td></tr>
<tr><td>Electric field due to volume charge
$\vec{E} = \int_v \frac{\rho_v dv}{4\pi\varepsilon}\frac{\vec{R}}{R^3} = \int_v \frac{\rho_v dv}{4\pi\varepsilon}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td> 
</td></tr>
<tr><td rowspan="4">Electric flux density
$\vec{D}$</td><td colspan="2">$\vec{D} = \varepsilon\vec{E} = \varepsilon_0\varepsilon_r\vec{E}$</td></tr>
<tr><td>Electric flux density due to line charge
$\vec{D} = \int_L \frac{\rho_L dl}{4\pi}\frac{\vec{R}}{R^3} = \int_L \frac{\rho_L dl}{4\pi}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td></td></tr>
<tr><td>Electric flux density due to surface charge
$\vec{D} = \int_S \frac{\rho_s ds}{4\pi}\frac{\vec{R}}{R^3} = \int_S \frac{\rho_s ds}{4\pi}\frac{\vec{r}-\vec{r'}}{\left|\vec{r}-\vec{r'}\right|^3}$</td><td></td></tr>
<tr><td>Electric flux density due to volume charge
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<tr><td>Electric flux
ψ</td><td colspan="2">$\psi = \int \vec{D}.\overrightarrow{ds} = \int \vec{D}.\hat{a}_n\, ds$</td></tr>
<tr><td>Gauss's law</td><td>$Q_{enc} = \psi = \oint_S \vec{D}.\overrightarrow{ds} = \int_v \rho_v dv$</td><td>$\vec{\nabla}.\vec{D} = \rho_v$</td></tr>
</table>

Electric potential (Voltage)	Electric potential due to point charge $V = \frac{w}{Q} = \frac{Q}{4\pi\varepsilon R} = \frac{Q}{4\pi\varepsilon\left\|\vec{r} - \vec{r'}\right\|}$ Electric potential due to line charge $V = \int \frac{\rho_L dl}{4\pi\varepsilon R} = \int \frac{\rho_L dl}{4\pi\varepsilon\left\|\vec{r} - \vec{r'}\right\|}$ Electric potential due to surface charge $V = \int \frac{\rho_s ds}{4\pi\varepsilon R} = \int \frac{\rho_s ds}{4\pi\varepsilon\left\|\vec{r} - \vec{r'}\right\|}$ Electric potential due to volume charge $V = \int \frac{\rho_v dv}{4\pi\varepsilon R} = \int \frac{\rho_v dv}{4\pi\varepsilon\left\|\vec{r} - \vec{r'}\right\|}$	
Electric field vs. Electric potential	$\vec{E} = -\vec{\nabla}V$	$\vec{\nabla} \times \vec{E} = \mathbf{0}$ For static electric field

7. MAGNETOSTATIC

Biot-Savart's law Magnetic field intensity	Magnetic field due to line current $\vec{H} = \int_L \frac{I\vec{dl} \times \vec{R}}{4\pi R^3} = \int_L \frac{I\vec{dl} \times (\vec{r} - \vec{r'})}{4\pi\left\|\vec{r} - \vec{r'}\right\|^3}$	
	Magnetic field due to surface current $\vec{H} = \int_S \frac{\vec{K}dS \times \vec{R}}{4\pi R^3} = \int_S \frac{\vec{K}dS \times (\vec{r} - \vec{r'})}{4\pi\left\|\vec{r} - \vec{r'}\right\|^3}$	
	Magnetic field due to volume current $\vec{H} = \int_v \frac{\vec{J}dv \times \vec{R}}{4\pi R^3} = \int_S \frac{\vec{J}dv \times (\vec{r} - \vec{r'})}{4\pi\left\|\vec{r} - \vec{r'}\right\|^3}$	
Ampere's law	$I_{enc} = \oint_L \vec{H}.\vec{dl} = \int_S (\vec{\nabla} \times \vec{H}).\vec{dS} = \int_S \vec{J}.\vec{dS}$	$(\vec{\nabla} \times \vec{H}) = \vec{J}$
Magnetic flux density	$\vec{B} = \mu\vec{H} = \mu_0\mu_r\vec{H}$	
Continuity equation	$\vec{\nabla}.\vec{J} = -\frac{\partial\rho_v}{\partial t}$	$\vec{J} = \sigma\vec{E}$
Magnetic flux	$\psi = \int_S \vec{B}.\vec{ds}$ (For one surface)	$\psi = \oint_S \vec{B}.\vec{ds} = 0$ $(\vec{\nabla}.\vec{B}) = 0$ (closed surface)